Test Report No. 2024 – 0517

Agricultural Machinery – Solar-Powered Irrigation System Brgy. Takdangan, Cabatuan, Iloilo

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Test Report Number	2024 – 0517
Date of Issuance	December 2, 2024
Valid Until	December 2, 2029
Agricultural Machinery	Solar-Powered Irrigation System
Type	Groundwater Source with Reservoir, Submersible Pumpset Powered by 9 Monocrystalline Photovoltaic Modules
Pump Brand	PEDROLLO
Pump Model	4SRM 13G/20-F-PD
Photovoltaic Solar Panel Brand	AE SOLAR
Photovoltaic Solar Panel Model	AE550MD-144
Test Requested by	KIRSKAT VENTURES Brgy. Bato, Bato, Mambusao, Capiz

SUMMARY

Performance Criteria	Requirement as per PNS/BAFS 324:2022	AMTEC Test
Maximum PV Performance Ratio	18[a]	25.28[b]
The pumpset shall have a test report with pump or performance curve prior to installation.	Provided	Not provided
Total volume discharge for whole day operation, m^3, minimum	22[c]	33.70

[a] Requirement for monocrystalline PV array.
[b] Data from 8:00 a.m. to 9:00 a.m. were not considered due to shading on the panels.
[c] Requirement based on system design.

OBSERVATIONS

1. Operator's manual	Provided for the controller/inverter only.
2. Previous test reports of similar brand and model	None
3. PV module manufacturer	AE ALTERNATIVE ENERGY GMBH Messerschmittring 54, 86343 Konigsbrunn, Germany
4. Pumpset manufacturer	SAN BONIFACIO (VR) Italy
5. Time of test	8:30 a.m. to 4:30 p.m.
6. Ease of cleaning the PV modules	The PV modules can be accessed for cleaning using a ladder or any platform.
7. Ease of adjusting and repairing of parts	All parts are accessible for adjustment and repair. However, the pump needs to be hoisted using a chain block to facilitate repairs.
8. Safety	Rotating parts, electrical wirings, and cables of the system are enclosed and properly sealed/coated.
9. Failure or abnormalities that may be observed on the system during and after the operation	No failure or abnormalities were observed on the system during and after the test operation.
10. The panels in the PV array should be installed with a rigid alloy rail made of non-corrosive material.	The PV array is installed with aluminum rails.
11. The mounting frame structure should be made from Galvanized Iron (GI) pipes or angular bars that are either primed, hot dipped galvanized with minimum of 5 mils or double coated with non-corrosive paint.	The mounting frame structure are composed of welded GI pipes and angular bars that were properly coated with paint.
12. A uniform type and specifications of PV modules, either monocrystalline or polycrystalline, shall be used for the whole array.	All monocrystalline PV modules used are of uniform specifications.
13. The PV modules shall be able to withstand a minimum gustiness and uplift of 180 kph. The standard degradation should be a minimum of 0.5% annually as per IEC 61215-1:2021 (Terrestrial photovoltaic (PV) modules – Design qualification and type approval – Part 1: Test requirements).	Not verified due to limited technical capabilities on site.

OBSERVATIONS (Cont'n)

14. The orientation of the PV array shall be as per site assessment. It is recommended that the PV array should be facing the south based on efficiency and performance.	The PV array is facing southeast.
15. The angle of inclination in degrees for the PV array shall be dependent on latitude of the location maximizing solar energy to be harvested.	The PV array 1 is inclined at 5.5°. The PV array 2 is inclined at 7.2°.
16. Placement of the PV array shall be away from any source of shades any time of the year. If applicable, the site of installation shall be cleared of any trees and other obstructions.	A nearby tree casted shadows on the PV modules starting from sunrise until 9:05 a.m. (Figure 6).
17. The spatial distance between each PV modules should be at least 20 mm.	The spatial distance between each PV module is 16.03 mm.
18. The type, capacity of pump, suction and discharge pipe material, and other basic components shall be dependent on the design of the SPIS.	The pump used is submersible pumpset. HDPE pipe was used for the discharge pipe
19. Dry running, overheating, overloading, voltage transient, and low/high voltage input protection shall be provided for the pumpset.	Low voltage input can be detected by the controller. Dry running, overheating, overloading, voltage transient, and high voltage input protection were not verified.
20. A trash rack, made of non-corrosive material or painted with protective coating, shall be installed along with the pump intake. The orientation of the pump shall be dependent on the system design (e.g. submersible or surface).	Not applicable for groundwater source.
21. The pumpset to be used shall be dependent on the water source and total head. Casings for groundwater sources shall be provided.	The system is equipped with a submersible pumpset with source from groundwater.
22. A discharge measuring device, such as flow meter or water meter, should be included with the pumpset.	Not provided
23. The pumpset controller and the PV array controller shall be insect-proofed and weather-proofed by using double-proof box and sealants.	The controller/inverter is installed inside a controller house located besides the PV modules. It was not verified if the controller was insect-proofed and water-proofed due to limited technical capabilities.

OBSERVATIONS (Cont'n)

24. A minimum of IP58 (splash proof) rating shall be used as per IEC 60529:2019 (Degrees of protection provided by enclosures [IP Code]).	Not verified due to lack of technical capabilities on site.
25. The solar panel controller should be encased with a combiner box.	Not applicable
26. The capacity of inverter shall be at least 25% higher than pumpset input power requirement.	The maximum output power if the inverter (2.2 kW) is 46.67% higher than the pumpset's input power requirement (1.5 kW).
27. The controller shall be fully automated based on manufacturers' specifications.	The controller/inverter is fully automated with a digital display.
28. Cables and wirings used for the SPIS shall be in accordance with the Philippine Electrical Code (PEC), using PV cables specific for PV modules. Wiring installations shall be properly protected for weather conditions and other external factors.	Electrical wirings and cables of the system are enclosed and sealed/coated. Wirings and cables for the submersible pumpset are encased in a PE pipe. However due to technical capabilities on site, it was not verified if the cables and wirings used for the SPIS are in accordance to the Philippine Electrical Code (PEC) and if the wiring installations are properly protected for weather conditions and other external factors.
29. If installed, the design of the reservoir should indicate that the intake be at least equal to the outflow during operation. The reservoir shall be able to withstand the design load. The water storage should also include a ladder, drain and overflow sensor, a freeboard, and an intake and outflow valves. Construction shall be in accordance with the PD 1096.	a. A 1000-liter tank is installed in the system. The inflow valve (D=38.10 mm) is smaller than the outflow valve (D=50.80 mm). b. The reservoir was able to withstand the load when it was at full capacity. The tank is provided with a ladder, drain, and intake and outflow valves. c. It was not verified if the construction of the reservoir conformed in accordance with PD 1096 due to limited technical capabilities on site.
30. A Rapid Shutdown Device (RSD) shall be installed for roof-mounted panels.	Not applicable
31. Lightning arrester shall be installed in accordance with Article 2.80 (Surge Arresters) of the National Electrical Code (NEC).	Not provided
32. Warning signs shall be installed for electrical components on-site.	Warning notices are provided on the PV modules and controller/inverter.

OBSERVATIONS (Cont'n)

33. Surge Protection Devices (SPD) shall be installed along with the PV module.	Surge protection device is integrated into the controller.
34. Personal Protective Equipment (PPE) shall be provided for each operator on site.	The operator was not provided with personal protective equipment during the test operation.
35. Other remarks	a. Markings and labeling are provided for the PV modules, and controller/inverter. b. The Test applicant insisted on performing the test even with intermittent pumpset discharge due to insufficient water supply from the source. c. The system stopped operating at 4:35 p.m. due to low irradiance.

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing
Date of Test	November 7, 2024
Location of Test	Brgy. Takdangan, Cabatuan, Iloilo 10°53'55.76" N, 122°29'4.11" E
Items Tested	Solar Panel Power Output, Input Power to the Pumpset, Pumpset Output Power, Discharge Capacity, Total Head, PV Performance Ratio, PV Module Temperature, and System Efficiency

METHODS OF TEST

The Solar-Powered Irrigation System was tested based on PNS/BAFS 325:2022 Solar-Powered Irrigation System – Methods of Test.

DESCRIPTION OF THE SYSTEM

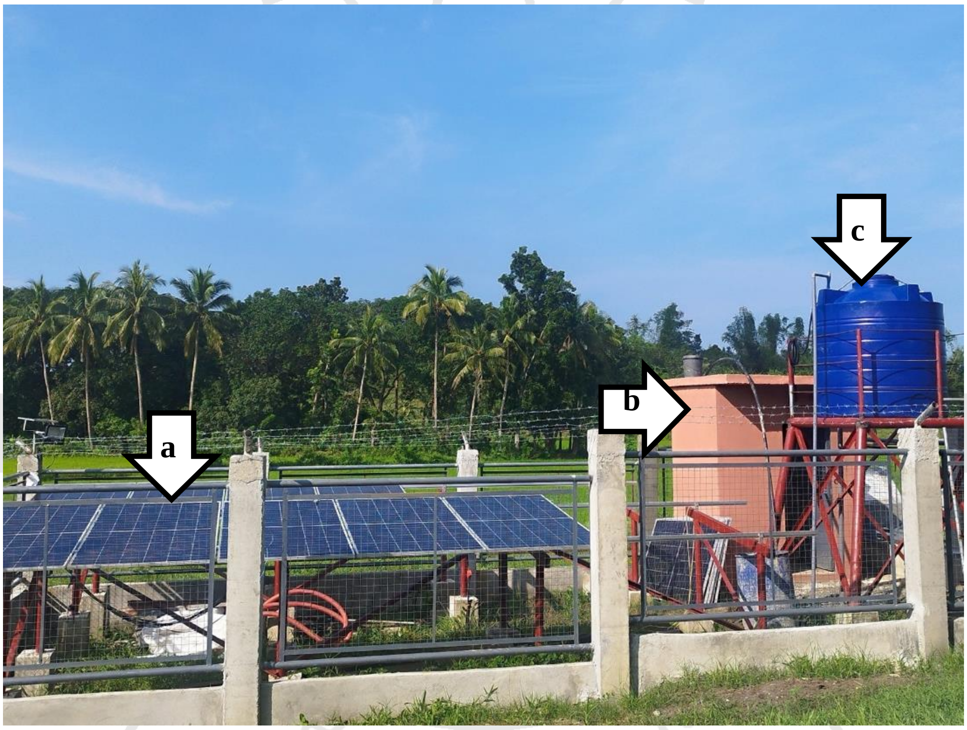

Figure 1. Different components of the Solar-Powered Irrigation System installed at Brgy. Takdangan, Cabatuan, Iloilo include the following: (a) PV modules, (b) Controller house, and (c) Storage tank.

DESCRIPTION OF THE SYSTEM (Cont'n)

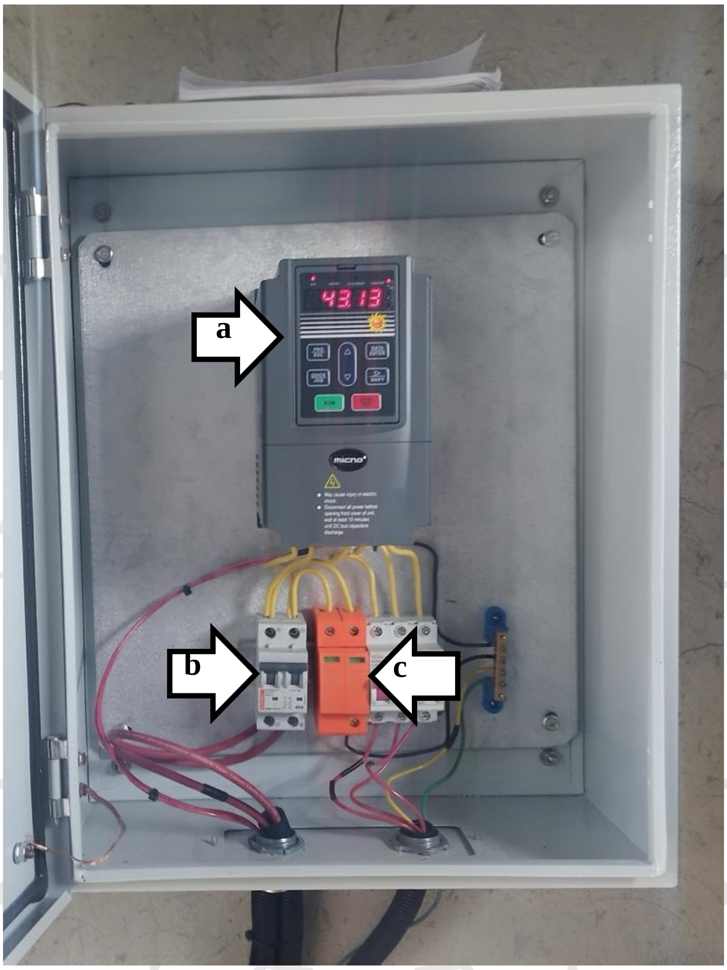

Figure 2. Different components of the Solar-Powered Irrigation System installed at Brgy. Takdangan, Cabatuan, Iloilo include the following:
(a) Controller/Inverter, (b) Circuit breaker, and
(c) Surge protection device

SPECIFICATIONS

Item		Manufacturer's Specification[a]	AMTEC Verification
1	Reservoir		
1.1	Overall dimensions, mm		
1.1.1	Diameter	ND	1400
1.1.2	Height	ND	1640
1.1.3	Height from the discharge outlet of the tank to the distribution channel	ND	2420
1.1.4	Thickness, mm	ND	NM[b]
2	Pumpset		
2.1	Pump		
2.1.1	Brand	PEDROLLO	PEDROLLO[a]
2.1.2	Model	4SRm 13G/20-F-PD	4SRm 13G/20-F-PD[a]
2.1.3	Serial number	221A8287P000188172	221A8287P000188172[a]
2.1.4	Type	Submersible, Centrifugal	Submersible, Centrifugal
2.1.5	Size, mm	ND	ND[c]
2.1.6	Maximum discharge, m^3/h	4.23	4.23[a]
2.1.7	Maximum total head, m	175	175[a]
2.1.8	Manufacturer	SAN BONIFACIO (VR) ITALY	SAN BONIFACIO (VR) ITALY[a]
2.1.9	Country of manufacture	Italy	Italy[a]
2.2	Electric motor		
2.2.1	Brand	PEDROLLO	PEDROLLO[a]
2.2.2	Model	4PD/2	4PD/2[a]
2.2.3	Type	Induction, Three-phase	Induction, Three-phase[a]
2.2.4	Rated power, kW	1.5	1.5[a]
2.2.5	Rated voltage, V	220	220[a]
2.2.6	Rated current, A	8.2	8.2[a]
2.2.7	Rated frequency, Hz	60	60[a]

[a] The data were obtained from the component's nameplate submitted by the test applicant.
[b] Not accessible for measurement.
[c] Not provided on the component/s nameplate.
Note: ND – No Data, NM – Not Measured

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification[a]	AMTEC Verification
3	Controller/Inverter		
3.1	Brand	MICNO	MICNO
3.2	Model	AE300-0104-2R2G-S2	AE300-0104-2R2G-S2
3.3	Enclosure class	ND	ND[b]
3.4	Input		
3.4.1	Maximum power, kW	ND	ND[b]
3.4.2	Maximum input voltage, DCV	440	440[a]
3.4.3	Maximum input current, DCA	ND	ND[b]
3.5	Output		
3.5.1	Maximum power, kW	2.2	2.2[a]
3.5.2	Maximum output voltage, ACV	220	220[a]
3.5.3	Maximum output current, A	17	17[a]
4	Solar module		
4.1	Brand	AE SOLAR	AE SOLAR
4.2	Model	AE550MD-144	AE550MD-144
4.3	Manufacturer	AE Alternative Energy GmbH	AE Alternative Energy GmbH
4.4	Country of manufacture	Germany	Germany
4.5	Cell type	Monocrystalline	Monocrystalline
4.6	Cell size, L × W, mm	ND	180 × 90
4.7	Cell configuration	ND	24 × 6

[a] The data were obtained from the component's nameplate.
[b] Not provided on the component nameplate.
Note: ND – No Data

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification[a]	AMTEC Verification
4.8	PV module dimension, mm		
4.8.1	Length	ND	2280
4.8.2	Width	ND	1130
4.9	Total number of solar modules	ND	9
4.9.1	Number of solar modules per string	ND	9
4.9.2	Number of strings	1	1
4.10	Nominal power, W_p, W	550	550[a]
4.11	Short circuit current, I_{sc}, A	13.67	13.67[a]
4.12	Open circuit voltage, V_{oc}, V	51.54	51.54[a]
4.13	Current at maximum power, I_{mpp}, A	12.92	12.92[a]
4.14	Voltage at maximum power, V_{mpp}, V	42.57	42.57 [a]
4.15	Reference standard temperature condition, T_{STC}	25	25[a]
4.16	Orientation	ND	Southeast
4.17	Angle of inclination, °	ND	PV array 1: 7.8 PV array 2: 5.5
5	Safety devices	ND	Circuit breaker, Surge protection device
6	Special features	ND	None

[a] The data were obtained from the component's nameplate.
Note: ND – No Data

RESULTS

Table 1. Performance of the Solar-Powered Irrigation System Installed in Brgy. Takdangan, Cabatuan, Iloilo.

Time of the Day	Solar Irradiance, W/m²	Solar Power, W	Module Tempe-rature, °C	Ambient Conditions		Solar Panel Output (DC)			Input to the Pumpset (AC)			Pumpset Output			PV Performance Ratio	System Efficiency, %
				Tempe-rature, °C	Relative Humidity, %	V	A	Power, W	V	A	Power, W	Total Head, m	Discharge Capacity, m³/h	Output Power, W		
8:30 a.m.	562.0	12916.11	39.2	28.5	81.3	336.0	3.2	1058.40	87.5	3.9	341.06	7.68	2.41	50.30	37.87	0.389
9:00 a.m.	670.6	15412.00	48.3	29.0	78.2	341.4	5.3	1809.16	143.1	5.6	801.08	8.88	3.80	91.98	19.24	0.597
9:30 a.m.	775.1	17813.66	47.2	29.7	78.1	333.9	7.2	2403.72	183.0	7.1	1299.30	9.80	4.62	123.23	13.71	0.692
10:00 a.m.	945.3	21725.26	51.4	30.1	79.3	366.5	8.6	3151.90	208.4	7.9	1646.36	10.57	5.22	150.17	13.20	0.691
10:30 a.m.	996.1	22892.00	61.5	31.9	74.9	363.9	8.7	3165.93	208.5	7.9	1647.15	10.45	5.13	145.86	13.90	0.637
11:00 a.m.	1056.6	24283.97	55.8	31.1	74.6	377.4	8.3	3132.42	208.5	7.9	1647.15	10.33	5.04	141.67	14.74	0.583
11:30 a.m.	984.4	22623.11	60.5	31.6	75.8	373.1	8.4	3134.04	208.4	7.9	1646.36	10.39	5.08	143.71	13.74	0.635
12:00 nn.	906.7	20838.14	49.6	31.5	72.1	381.2	8.3	3163.96	208.4	7.9	1646.36	10.31	5.02	141.13	12.66	0.677
12:30 p.m.	330.3	7591.09	46.7	31.1	77.4	339.5	5.1	1731.62	143.6	5.4	775.44	8.70	3.62	85.88	9.79	1.13
1:00 p.m.	869.8	19989.33	61.9	31.9	74.1	363.0	8.6	3121.80	208.6	7.9	1647.94	12.13	5.10	168.51	12.13	0.843
1:30 p.m.	781.3	17956.15	58.0	32.4	72.4	347.3	8.6	2987.07	206.2	7.8	1608.36	10.35	5.05	142.32	11.16	0.793
2:00 p.m.	304.9	7006.57	55.7	32.8	72.5	329.5	4.9	1614.55	130.7	5.3	692.71	8.51	3.42	79.31	10.11	1.13
2:30 p.m.	805.9	18521.52	57.9	33.0	72.5	331.5	8.2	2718.30	191.6	7.3	1398.68	9.93	4.72	127.75	13.24	0.690
3:00 p.m.	574.9	13213.35	53.4	33.6	66.3	329.7	5.7	1879.29	149.9	5.8	869.42	9.00	3.92	95.97	15.20	0.726
3:30 p.m.	444.2	10208.02	48.4	33.8	68.3	333.6	4.6	1534.56	123.1	5.0	615.50	8.34	3.25	73.79	16.58	0.723
4:00 p.m.	297.7	6841.86	43.3	33.7	66.6	330.7	5.3	1752.71	85.8	3.8	326.04	7.65	2.37	49.40	20.98	0.722
4:30 p.m.	190.1	4368.65	39.2	33.7	70.8	326.5	2.5	816.25	59.6	2.9	172.84	7.24	1.67	32.88	25.28	0.753

RESULTS (Cont'n)

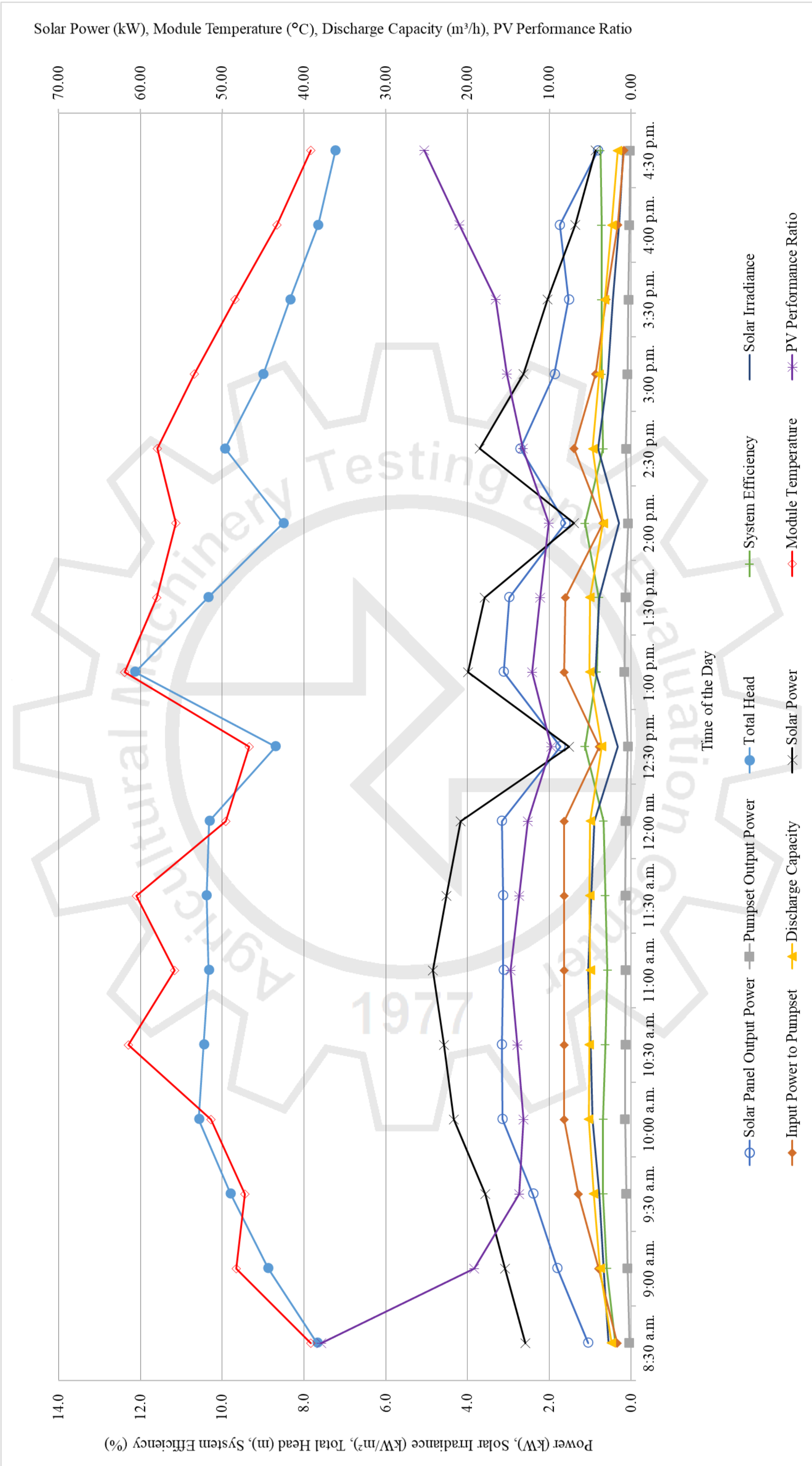

Figure 3. Performance Curves of the Solar-Powered Irrigation System Installed in Brgy. Takdangan, Cabatuan, Iloilo.

OBSERVATIONS

Figure 4. Nameplate of the controller/inverter.

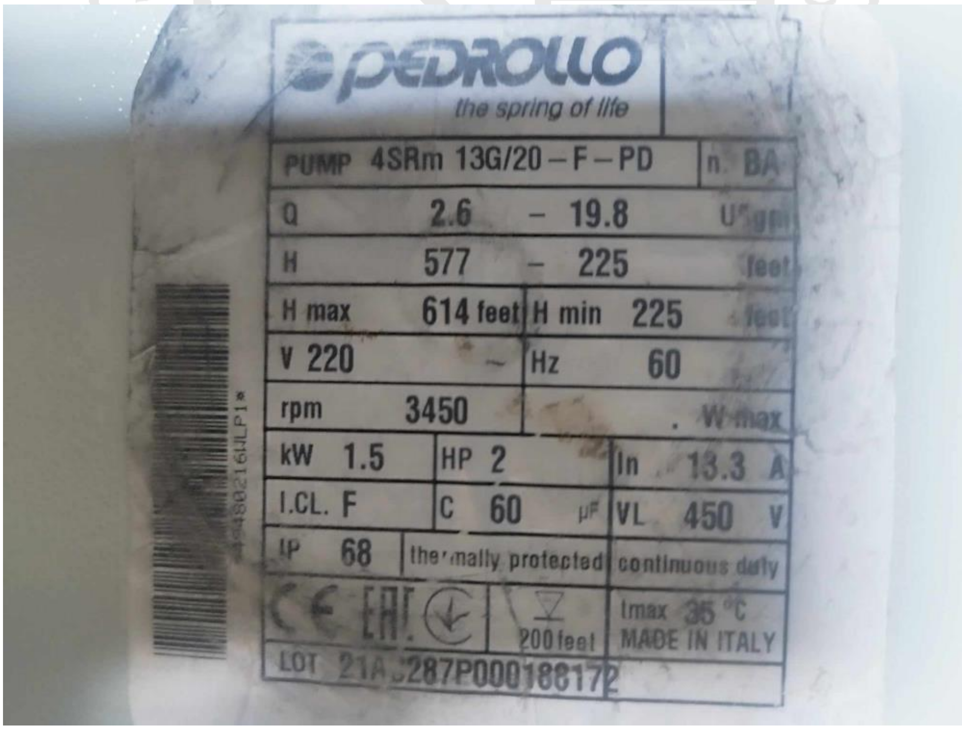

Figure 5. Nameplate of the pumpset as submitted by the test applicant.

OBSERVATIONS

Figure 6. Shading on the panels.

OBSERVATIONS (Cont'n)

Figure 6. Nameplate of the PV module.

Tested by:

FRANKLIN C. PAJARON
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DEO-JAY T. MANGLAL-LAN
Test Engineer

Approved for Release:

ARTHUR L. FAJARDO
AMTEC Director

0 9 DEC 2024
Date

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